Notes: Hendricks, Piccione, and Tan (Econometrica, 1999)
Equilibria in Networks

Contents

1 Big picture

- **Question.** Why do hub-spoke networks emerge when carriers compete over networks and then compete for travelers?
- **Core setting.** Two carriers choose networks over n cities. After networks are chosen, profits in each city-pair market depend on the lengths of the paths offered by the two carriers.
- **Economic force.** Economies of density make point-to-point service unprofitable. A direct connection must carry travelers from many origin-destination markets to cover fixed costs.
- **Aggressive competition result.** Under Bertrand-like competition, a complete hub-spoke carrier can deter entry. A monopoly hub-spoke network is an equilibrium outcome.
- **Soft competition result.** If carriers earn profits even when neither has a path-length advantage, competing hub-spoke networks can be equilibria.
- **Takeaway.** Hub-spoke networks depend on how network cost savings interact with market power and the intensity of downstream competition.

2 Objects and measurement: networks, paths, and profits

A network is an undirected graph. Carrier i's network is $X_i(g,h) = 1$ if cities g and h are directly connected. Its size is

$$m_i = \frac{1}{2} \sum_{g,h \in N} X_i(g,h).$$

A hub-spoke network is a network in which all direct connections share one hub city. A complete hub-spoke network has $n-1$ links.

A connection function $\tau_i(g,h)$ gives the shortest path length in carrier i's network, or ∞ if the city pair is not connected. Operating profit in a market is $\pi(z_i, z_j)$, and monopoly profit is $\pi_M(z) = \pi(z, \infty)$. Network profit is

$$\Pi_i(\tau_i, \tau_j) = \sum_z \sum_{(g,h) \in \Gamma_i(z)} \pi(z, \tau_j(g,h)) - F(\#\Gamma_i(1)/2).$$

Interpretation: The payoff depends on all origin-destination markets induced by the network, not just direct links.

3 Models and equilibrium specifications

3.1 Baseline assumptions

- A1: shorter own paths weakly increase profits: $\pi(z,y) \geq \pi(z+1,y)$.
- A2-i: point-to-point monopoly is not profitable: $F(n(n-1)/2) > n(n-1)\pi_M(1)$.
- A2-ii: complete hub-spoke monopoly is profitable: $2(n-1)\pi_M(1) + (n-1)(n-2)\pi_M(2) > F(n-1)$.
- A3: duopoly profits are bounded by monopoly profits.

3.2 Aggressive competition

$$\text{BC: } \pi(z,y) = 0 \quad \text{whenever } z \geq y.$$

Interpretation: A carrier earns positive profits only if it offers a strictly shorter path.

3.3 Soft competition and modularity

$$\text{A4: } (n-2)(n-3)\pi(2,2) > F(n-2).$$

Quasi-submodularity means mismatching path lengths is relatively attractive; quasi-supermodularity means matching path lengths is relatively attractive.

3.4 Ruling out nonhub equilibria

$$\pi(1,z) - \pi(2,z) < \inf_s \{\pi(2,s) - \pi(3,s)\}, \qquad \pi(1,z) > \pi(2,z).$$

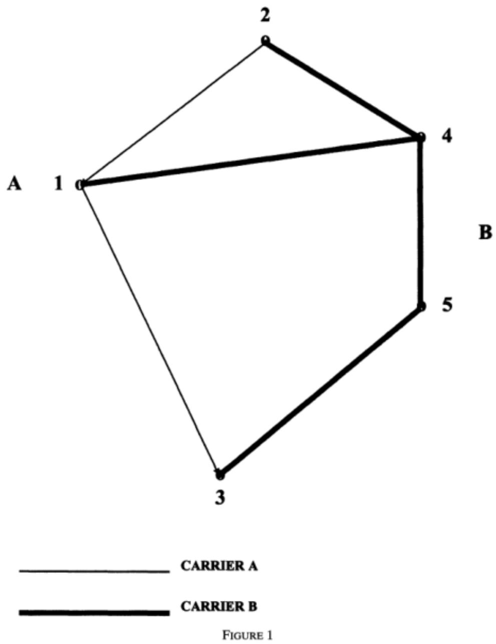
4 Equilibrium logic and theorem map

Result	Conditions	Equilibrium implication
Theorem 1	A1–A3 and BC	No active hub-spoke duopoly; one complete hub-spoke carrier plus empty rival is an equilibrium.
Theorem 2	Path length costless limit	Equilibria are one empty carrier and one tree; perturbations select hub-spoke trees.
Theorem 3	A1–A4 and modularity	Complete hub-spoke duopoly equilibria exist.
Theorem 4	A1–A5, $n > 5$, $F(m) = mf$	Every pure-strategy equilibrium network is hub-spoke.

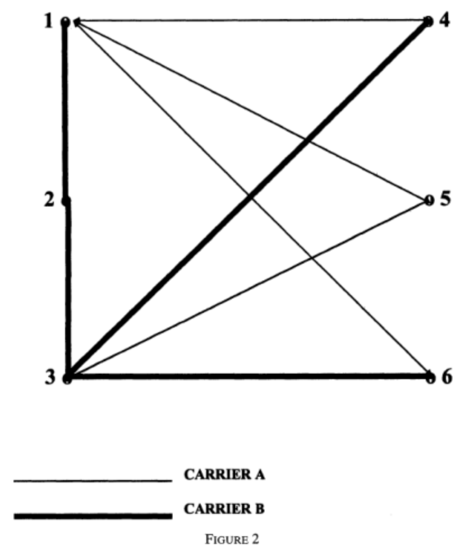
5 Tables and figures

The paper has no empirical tables. Its original visual material is a sequence of network diagrams. The figures below are directly cropped from the paper and grouped compactly.

5.1 Figure 1 + Figure 2: nonhub duopoly examples



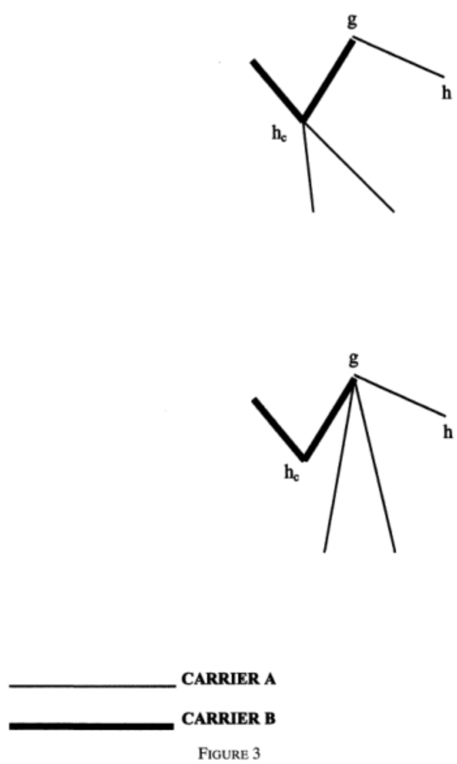
(a) Figure 1: five-city nonhub duopoly equilibrium.



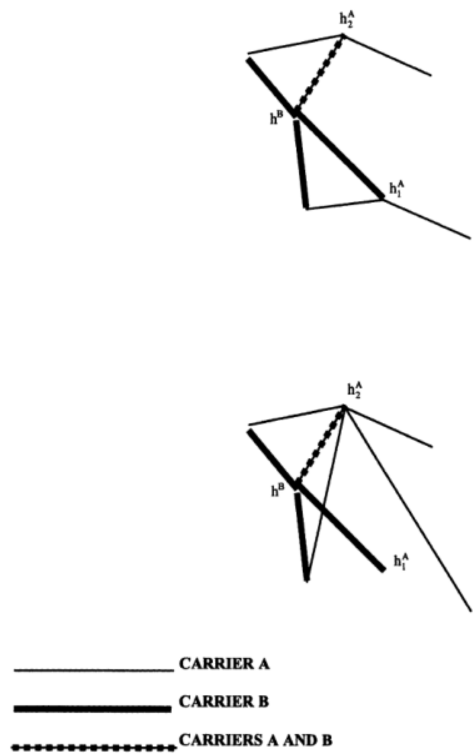
(b) Figure 2: six-city nonhub equilibrium.

Read: These examples show that nonhub duopoly equilibria can exist when carriers strategically position networks to avoid head-to-head competition or preserve path-length advantages.
Economic message: Hubbing is not inevitable; nonhub networks can soften competition.

5.2 Figure 3 + Figure 4: proof transformations



(a) Figure 3: transformation around a shared/nearby hub.



(b) Figure 4: transformation with disjoint hubs.

Read: The proof identifies local rewiring moves that improve profits and therefore rule out candidate nonhub best replies.
Economic message: Once long paths are costly enough, nonhub structures are vulnerable to profitable rewiring toward hub-spoke form.

5.3 Figure 5: combining components

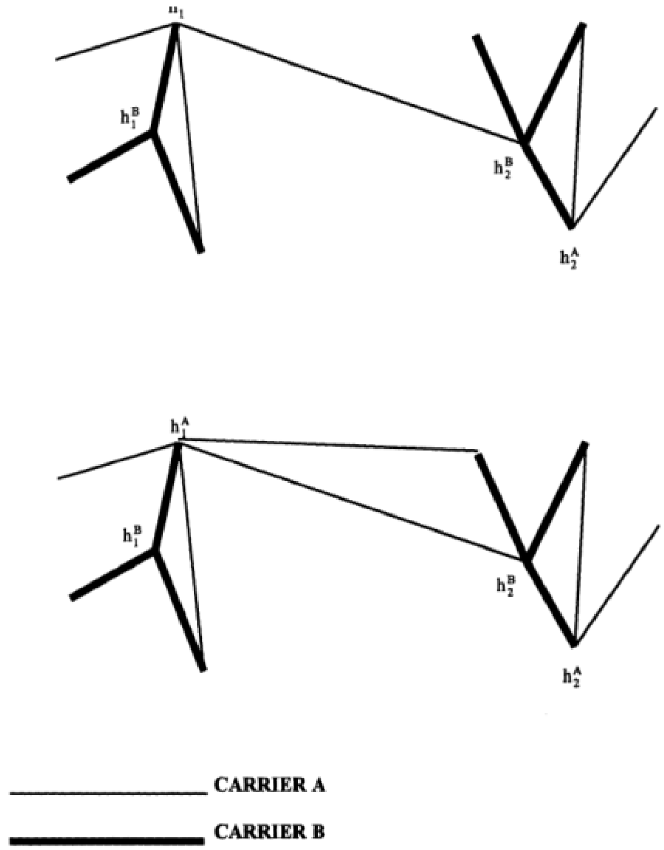


Figure 3: Figure 5: connecting separate components creates new markets.

Economic message: One extra link between components can create many new connected markets, so fragmented hub-spoke pieces tend to be merged.

6 Short summary

- The paper models network choice as a two-stage game: choose networks, then compete for travelers.
- Economies of density make point-to-point networks unprofitable but complete hub-spoke networks profitable.
- Under aggressive competition, a complete hub-spoke carrier can deter entry.
- Under softer competition, competing hub-spoke networks can be equilibria.
- Nonhub equilibria can exist when they soften competition or create local length advantages.
- Strong aversion to multi-stop paths restores hub-spoke structure in all pure-strategy equilibria.

7 Conclusion

Hub-spoke networks emerge from the interaction of density economies, path-length preferences, and downstream competitive intensity. Strategic competition does not mechanically select hub-spoke networks; in some environments it can support nonhub equilibria. The paper's contribution is to characterize when hub-spoke networks are equilibrium outcomes and when deviations can survive.